# Policy and information news in German consumer-goods production

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11 October 2026

**Abstract**

I examine how separating the current policy and information contributions to ECB announcements changes the estimated durability contrast in German consumer-goods production. Matched local projections on 308 monthly origins give a twelve-month policy-component contrast of −2.29 percentage log-change points per ten-basis-point factor contribution, compared with −1.03 for the recomposed surprise. An exact decomposition links their 1.26 gap to the component-response ranking and the controlled announcement mixture, retaining joint uncertainty. Prespecified conservative 95 per cent bounds are $[-0.11, 3.31]$; every declared sensitivity also includes zero. A post-estimation zero-product test gives nominal $p = 0.036$ for the primary construction, while joint-region bounds and the construction and period alternatives leave a stronger conclusion unresolved. A durable-stock benchmark illustrates user-cost and demand mechanisms, while production measurement and relative-price feedback limit a direct household interpretation. The point estimates indicate that combining announcement contributions weakens the estimated durability contrast without requiring a positive relative information response. The declared procedure leaves the population adjustment unresolved, and a causal transmission interpretation requires additional measurement and orthogonality assumptions.

*Keywords:* monetary policy; information shocks; consumer-goods production; local projections

*JEL classification:* C22, E32, E52.

## 1 Introduction

Monetary-policy announcements can change expectations about both financing conditions and the economic outlook (Jarociński and Karadi, 2020). An unexpected rise in near-term rates may signal tighter policy, favourable information about activity, or both. I ask how separating these current announcement contributions changes the estimated response of German durable relative to non-durable consumer-goods production. The economic problem is that a smaller coefficient on combined news need not imply weaker policy transmission: the measured announcement mixture can change the apparent durability contrast.

Durability makes this comparison economically meaningful. A stock supplies services after purchase, allowing replacement expenditure to adjust more than current services. Its user cost depends on financing, expected resale prices and risk; income and wealth news can also change purchases. These mechanisms can produce different relative responses to policy and information contributions, but relative prices and firms' financing can alter the ordering (Barsky et al., 2007; Durante et al., 2022; Dedola and Lippi, 2005). German production additionally serves exports and inventories. I therefore use household theory to organise possible transmission mechanisms while estimating a defined production contrast.

I compare projections on separated contributions with a projection on their recomposed rate factor, holding production outcomes, origin months, controls and rate units constant. Both already condition on decomposed announcement histories, so the comparison isolates the current contribution conditional on those histories. The primary object is the recomposed coefficient less the policy-component coefficient at twelve months. Established partial-regression algebra decomposes this gap into a controlled information

weight and the component-response difference. The weight is a covariance ratio, not an announcement probability. Its sampling variation and covariance with the response difference enter population inference.

The policy-component point contrast is −2.29 percentage log-change points per ten-basis-point contribution to the supplied factor, compared with −1.03 for the recomposed surprise. The 1.26 gap moves the point response towards zero, but its prespecified conservative 95 per cent bounds $[-0.11, 3.31]$ include zero. Every declared twelve-month sensitivity retains a positive point gap and an interval including zero; the earlier-period policy-contrast interval also includes zero. I retain the declared procedure and these adverse findings. Its non-rejection is specific to the chosen conservative construction, rather than a claim that every valid zero-gap test would agree.

My project-specific argument concerns the ranking of relative component responses and the mixture in which announcement news is measured. A positive adjustment does not require information news to produce relative expansion: a less negative information response can also shift the combined coefficient towards zero. The earlier-period point estimates illustrate this distinction. The matched comparison assesses this measurement effect without treating a change in a projection coefficient as evidence of a changed causal mechanism. Existing research establishes durability heterogeneity and information-adjusted monetary shocks; the contribution here is their focused economic reassessment on the same defined outcome and design. Section 2 develops the mechanism and literature comparison; Sections 3–5 define measurement, estimands and inference; Sections 6–7 assess the evidence and its economic meaning.

# 2 Economic rationale and related evidence

## 2.1 Durability, purchases and production

A durable purchase adds to a stock that supplies services over time; expenditure and service consumption are distinct objects (Mankiw, 1982; Barsky et al., 2007). I use a simple benchmark to separate user-cost and demand channels, rather than to calibrate a German production elasticity. With non-durable consumption $C_t$ as numeraire, end-of-period stock $S_t$, purchases $X_t$ and depreciation $0 < d \leq 1$,

$$S_t = (1 - d)S_{t-1} + X_t. \tag{1}$$

For an interior household with logarithmic utility weights $\alpha$ and $1 - \alpha$, the stock-service condition is

$$\frac{S_t}{C_t} = \frac{1 - \alpha}{\alpha q_t}, \qquad q_t = p_t - (1 - d)\mathbb{E}_t[M_{t+1}p_{t+1}] > 0. \tag{2}$$

Here $p_t$ is the durable price in non-durable units and $M_{t+1}$ the household's marginal-utility discount factor. User cost deducts the risk-adjusted retained resale value from today's price. Holding current and expected relative prices and the risk-adjustment term fixed, a higher real financing rate raises user cost when $d < 1$; the financing effect vanishes at $d = 1$. Expected resale prices and risk can offset the cost movement, while income and wealth can change the purchase response through $C_t$. Appendix C.1 derives the conditions and states the required interiority, resale and timing assumptions.

At a stationary-stock base with positive replacement purchases $\bar{X} = d\bar{S}$, holding inherited stock fixed gives the local relation

$$\mathrm{d}\log(X_t/C_t) = \left(\frac{1}{d} - 1\right)\mathrm{d}\log C_t - \frac{1}{d}\mathrm{d}\log q_t. \tag{3}$$

The stock-flow ratio $1/d$ amplifies the purchase adjustment needed to change services. Conditional on current non-durable consumption, a user-cost increase reduces the relative purchase flow; favourable demand or wealth news can act in the opposite direction when $d < 1$. These are local partial responses. Adjustment costs, borrowing constraints and endogenous relative prices can alter them. In the general-equilibrium models of Barsky et al. (2007, pp. 990–992, 995), which sector has sticky prices can even

reverse durable co-movement with a monetary expansion. Their long-lived-durable argument does not supply an unconditional ordering for these production groups.[1]

The mechanism motivates heterogeneity without predicting the sign of the information-component durable-minus-non-durable response. Favourable news can raise both sectors equally, or raise non-durable activity more. A positive adjustment gap requires an information response above the policy response when the weight is positive; it does not require a positive relative information response. Durante et al. (2022) provide firm-level evidence of stronger investment and sales responses among durable producers, while the household benchmark explains one reason the purchase flow can be sensitive.

That mechanism does not translate directly into a household-demand elasticity estimated from production indices. German production serves both domestic and foreign purchasers. Inventory accumulation or depletion can separate the timing of sales from the timing of production. Firms' financing requirements can also change their capacity to maintain output independently of a household's decision to postpone a purchase. Dedola and Lippi (2005) relate industrial responses to financing requirements, borrowing capacity and firm size as well as durability. The relative production coefficient therefore combines potentially different demand, financing and adjustment responses. I use durability to motivate the comparison while keeping these alternative explanations within its interpretation.

The classification makes this distinction material. I use Eurostat's official consumer-goods production groupings, which allocate industries at both division and group levels. Motor-vehicle manufacturing belongs to capital goods under the applicable classification, even though household purchases of cars can enter durable expenditure.[2] The non-durable production grouping likewise excludes household services. I therefore cannot transfer a result for broad durable manufacturing, investment goods or household consumption to these two production categories without checking what the comparison actually measures.

The closest published production studies already establish heterogeneous monetary transmission. Peersman and Smets (2005) study eleven industries in seven euro-area countries, including Germany, over 1980–1998. Their durability classification includes capital and intermediate durables. Dedola and Lippi (2005) examine twenty-one manufacturing sectors in five countries, including Germany, with domestic-policy identification. Choi et al. (2024) subsequently use an annual international industry panel and interactions between monetary shocks and industry characteristics.[3] Their differential approach also recognises that a common macroeconomic disturbance can have unequal sectoral effects. Subtracting the two production outcomes removes an identical common response, but does not remove a disturbance simply because it occurs in both sectors.

## 2.2 Information adjustment and the economic comparison

An interest-rate surprise around a policy announcement can convey more than an unexpected change in the policy stance. Under the interpretation developed by Jarociński and Karadi (2020), policy tightening raises the rate surprise while lowering equity prices, whereas favourable central-bank information can raise both. The latter component may affect expected activity differently from the policy component. A regression on their combined rate contribution can consequently describe a different production response from a regression that separates them. In this study, the same supplied factor provides the rate units for both comparisons; its contribution is not an equal-sized change in a particular ECB policy rate.

The component labels remain identifying interpretations. Bauer and Swanson (2023a) show that surprises can reflect an unexpectedly strong systematic response to public economic news while retaining conventional rate–equity signs. This complicates both private-information interpretations and exogeneity

[1] The simple US expenditure dynamics tested by Mankiw (1982) are rejected; I use its stock-service distinction as a conceptual foundation, not empirical validation. The benchmark's real return is period-matched to depreciation. The empirical OIS factor is nominal and annualised, so inserting its ten-basis-point contribution into the household derivative would require an additional mapping.

[2] The production allocation is specified in Commission Implementing Regulation (EU) 2020/1197, Annex II.A. The supply-use manual includes motor vehicles in household consumer durables (Eurostat, 2008, pp. 151, 397). These are different measurements.

[3] For Choi et al., detailed methods consultation uses the December 2023 Bank of England Working Paper 1024 and the published preview, not the final full methods text.

of the policy-labelled component. Their evidence concerns Federal Reserve announcements and does not measure the magnitude of this issue in the German application. It nevertheless limits what rate-equity co-movement establishes. Their published monthly analysis also demonstrates why the information content of high-frequency surprises matters for subsequent macroeconomic estimation (Bauer and Swanson, 2023b). I examine sensitivity to the adopted decomposition without identifying the information contribution as superior private knowledge or its response as a separately measured expectations mechanism.

Information-adjusted durability comparisons are also established. Durante et al. (2022) combine firms from four large euro-area economies, including Germany, with the original poor-man ECB proxy. Their annual investment and sales responses concern broader manufacturing groups. Casalis and Krustev (2022) compare quarterly durable and non-durable expenditure, including services in the latter, using a monetary-conditions shock that encompasses the broader lending environment.[4] These comparisons motivate heterogeneity while differing in outcome, frequency, categories and intervention from the matched production comparison here.

The close Flaccadoro and Ropele (2026) combines monthly euro-area consumer-goods production, information-adjusted ECB shocks and local projections. Its broad-durability comparison and coarse two-digit consumer-group mapping differ from my detailed official allocation. This unpublished June 2026 supplement rules out treating that combination of methods as a contribution by itself. I focus on the matched current-announcement gap conditional on shared histories, rather than claim historical priority for the exact contrast.

## 3 Data and measurement

I use German monthly production-volume indices for durable and non-durable consumer goods from Eurostat's `sts_inpr_m` dataset. Both indices are seasonally and calendar adjusted and referenced to 2021=100. They measure production in the official main industrial groupings (MIGs). These groupings combine NACE Rev. 2 divisions and groups, rather than assigning every two-digit manufacturing industry to a durability category (European Commission, 2020; Eurostat, 2026). Consumer durables comprise consumer electronics, optical and photographic equipment, domestic appliances, miscellaneous transport equipment, furniture, jewellery and musical instruments. Crucially, motor-vehicle manufacturing belongs to capital goods.[5] I therefore interpret the outcome as relative production in these defined manufacturing groupings. Exports, inventories and financing can influence production alongside domestic household demand; the indices do not separately identify these channels.

My price control uses Germany's unadjusted all-items HICP, `prc_hicp_midx`, in the historical ECOICOP classification, referenced to 2015=100. I construct annual logarithmic inflation as

$$\pi_t = 100 \log(P_t / P_{t-12}) \,. \tag{4}$$

The series ends in December 2025, but this does not restrict the primary response window. At the last primary origin, August 2025, the latest lagged price observation is July 2025. Future production is available through August 2026; future prices do not enter the control matrix. I retain a coherent historical price series throughout.[6] Constant changes in index reference years cancel from the logarithmic changes used here.

The announcement variables are an author-maintained update of the policy and information decomposition introduced by Jarociński and Karadi (2020), using the EA-MPD monetary-event window of Altavilla et al. (2019). I fix the update to its 7 January 2026 revision (Jarociński, 2026). Its near-term rate

[4] For the consumption data and identification I consulted ECB Working Paper 2386, April 2020, alongside the published abstract and introduction; the final journal methods text was not consulted.

[5] The durable grouping contains NACE Rev. 2 groups 26.4, 26.7, 27.5 and 30.9, division 31, and groups 32.1 and 32.2. Division 29, including motor vehicles, trailers and semi-trailers, is allocated to capital goods in Regulation 2020/1197, Annex II.A. Short-term statistics retain NACE Rev. 2 through 2027.

[6] Eurostat has archived the historical classification and publishes a full-series ECOICOP version 2 successor, `prc_hicp_minr`. I do not splice these classifications. Calendar-month indicators accommodate a fixed seasonal pattern in the regressions; they do not turn HICP into an officially seasonally adjusted series.

factor is the first principal component of one-, three- and six-month and one-year OIS surprises. The four inputs are standardised by their sample standard deviations, without subtracting their sample means before component extraction. The component is then scaled to the standard deviation of the one-year OIS surprise and expressed in annualised percentage points. The accompanying equity surprise is 100 times the logarithmic change in the Euro Stoxx 50.

I use the supplied median admissible rotation of this rate–equity system. A common selected angle applies across announcements and permits policy and information contributions within the same event (Jarociński, 2022, 2026). Their sum reconstructs the rate factor. This is a later point-selected implementation, rather than the original article's monthly implementation.[7] Event-column orthogonality is not preserved automatically by monthly aggregation, control residualisation or sample trimming.[8] Consequently, it cannot justify suppressing joint policy–information dependence in the empirical comparison. I divide both contributions by 0.10, so a unit change represents ten basis points of the normalised factor contribution. This normalisation does not equate the estimates with the effect of an actual ten-basis-point ECB policy-rate increase.

I use the supplied monthly CSV contributions unchanged. Independent aggregation of the retained event columns reproduces them to the recorded numerical precision. I preserve supplied zero months and leave observations outside source coverage missing. After twelve lagged histories and endpoint requirements, the primary twelve-month comparison contains 308 origins, January 2000 to August 2025. The response path uses 296 common origins, ending August 2024. The fixed 10 October 2026 retrieval incorporates revised historical observations, rather than reconstructing information available at each announcement. Appendix B records source coverage and reconciliation. These checks establish measurement consistency; announcement identification and the interpretation of the selected components remain substantive assumptions.

Table 1 puts the reporting scale in context. A ten-basis-point factor contribution is approximately three sample standard deviations of monthly policy news, rather than a typical announcement-size assumption. The primary calendar retains 45 months without source events. Durable and non-durable monthly growth have sample standard deviations of 3.85 and 1.89 percentage log-change points, respectively; these describe production variability, rather than policy responses.

Table 1: Monthly announcement variation on the primary origins

| Contribution | Sample SD, basis points | Control-residual SD, basis points |
|---|---|---|
| Policy contribution | 3.32 | 2.77 |
| Information contribution | 2.79 | 2.41 |
| Recomposed factor | 4.35 | 3.59 |

January 2000–August 2025, $n = 308$, including 45 months without retained events. Both columns use divisor $n - 1$. The right column removes the same 72 controls defined in Section 4 from each current contribution, without removing the other current component. These are descriptive standard deviations in normalised factor units, not disturbance standard errors or actual policy-rate changes.

# 4 Matched empirical design

## 4.1 Outcome and announcement units

Let $D_t$ and $N_t$ denote strictly positive durable and non-durable production indices. For each horizon $h$, I use

$$Y_{t,h} = 100 \left\{ \log \frac{D_{t+h}}{N_{t+h}} - \log \frac{D_{t-1}}{N_{t-1}} \right\}. \tag{5}$$

[7] For the later construction, the consulted version of Jarociński (2022) is the author manuscript dated 14 September 2022, §§2.1–2.2 and Appendix B; I did not consult its final publisher text.

[8] The code's orthogonality concerns uncentred column products; it does not establish zero centred covariance or independence.

This is the difference between the two cumulative percentage log changes, rather than a ratio of physical production quantities. It is invariant to separate constant rebasing of the indices. At $h = 12$, the endpoint is one year after the announcement month relative to the preceding month, so the outcome sums thirteen monthly changes. I attribute the local-projection approach to Jordà (2005); the inference choices below rely on the consulted methods of Stock and Watson (2018) and Montiel Olea and Plagborg-Møller (2021).[9]

Write $m_t = m_t^{\text{raw}}/0.10$ and $c_t = c_t^{\text{raw}}/0.10$ for the supplied policy and information contributions, and recompose $r_t = m_t + c_t$. All three regressors share rate-factor units. A unit is a ten-basis-point contribution to the normalised factor, not an observed policy-rate intervention of that size. I do not independently standardise the two components, since that would alter the comparison's units and its decomposition.

## 4.2 Controls, sample and comparison

I estimate the two ordinary least-squares projections

$$Y_{t,h} = \beta_{m,h} m_t + \beta_{c,h} c_t + Z_t^\top \eta_h + e_{t,h}, \tag{6}$$

$$Y_{t,h} = \gamma_h r_t + Z_t^\top \zeta_h + v_{t,h}. \tag{7}$$

The same rows and $Z_t$ enter both. They define linear-projection coefficients; the equations do not, by themselves, establish a correctly specified conditional mean or a structural intervention. A structural reading needs more. In a noiseless benchmark, let the supplied components $s_t = (m_t, c_t)^\top$ measure latent innovations $\varepsilon_t$ through a constant invertible matrix $A$, and let the structural relative response be $b_h$. Conditional orthogonality of the remaining production disturbance to the measured components then gives

$$s_t = A\varepsilon_t, \qquad \beta_h = A^{-\top} b_h, \quad \beta_h = (\beta_{m,h}, \beta_{c,h})^\top. \tag{8}$$

Appendix C.2 proves this relation. A known diagonal mapping recovers the separately scaled latent responses. An unknown dense mapping mixes them; measurement noise generally breaks this identity. Orthogonal observed columns do not establish a diagonal mapping, exogeneity or response invariance. Rate–equity restrictions select an admissible interpretation of announcement news, rather than prove these conditions. I therefore estimate supplied-component projections and state the stronger assumptions needed to call them causal responses (Jarociński and Karadi, 2020; Stock and Watson, 2018).

The controls comprise an intercept, eleven calendar-month indicators and twelve lags of five histories: durable monthly log growth, non-durable monthly log growth, annual German HICP log inflation, and both scaled announcement components. I use preceding reference months, without contemporaneous production or inflation controls. Twelve lags allow a year of preceding dynamics, while the six-lag check assesses the parameter cost on unchanged dates. Month indicators allow residual calendar effects and announcement-opportunity patterns. These finite histories define the projection rather than demonstrate a sufficient information set or a maximum dependence length. Both models already condition on decomposed announcement histories: the primary gap isolates separation of the current contribution, conditional on those histories. It does not measure the total effect of abandoning information adjustment at every date.

The full primary regression has 74 columns and 234 residual degrees of freedom. Every origin requires its contemporaneous components, complete lagged controls, and positive production at $t - 1$ and $t + h$. Future prices and shocks are not required. No missing observation is imputed and no primary pandemic origin is excluded. At $h = 12$, the last available production endpoint implies August 2025 as the final origin, producing 308 monthly observations from January 2000. This is a historical projection over different monetary-policy regimes and pandemic states. A stable structural response across them needs stronger assumptions than the pooled specification.

I fixed the primary horizon, sample, outcome, units, controls and interval procedure before fitting the response models. Appendix B.1 records chronology and the prespecified checks. The direct use of

[9] I consulted Jordà's publisher metadata and abstract, rather than its original full methods text. The latter two sources were consulted in their published versions. For the lag-augmentation result I also consulted the authors' January 2026 correction (Montiel Olea and Plagborg-Møller, 2026); I do not presume that my growth-control design inherits that theorem.

supplied contributions has no IV first stage. Small residual variation or influential announcements concern precision and numerical regularity here; they are not weak-instrument statistics (Stock and Watson, 2018).

The primary adjustment gap and the component-response difference are

$$\delta_h = \gamma_h - \beta_{m,h}, \qquad \theta_h = \beta_{c,h} - \beta_{m,h}. \tag{9}$$

The policy coefficient is a companion estimand. Neither its individual interval nor the information coefficient's interval replaces an interval for $\delta_h$.

**Proposition 1** (Matched rate-surprise comparison)**.** *On identical rows, controls and units, let $Z$ have full column rank, $[m, c, Z]$ have full rank and $n >$ rank$[m, c, Z]$. Then*

$$\widehat{\delta}_h = \widehat{w}_{c,h}\widehat{\theta}_h, \qquad \widehat{w}_{c,h} = \frac{r^\top M_Z c}{r^\top M_Z r}, \qquad M_Z = I - Z(Z^\top Z)^{-1}Z^\top. \tag{10}$$

This application of the partial-regression result reviewed by Ding (2021) is proved in Appendix A.1. Its economic interpretation is straightforward. If the information weight is positive and the information response exceeds the policy response, combining the components shifts the estimated coefficient upward. When the policy response is negative, this makes the relative contraction appear smaller. But the weight need not lie in $[0, 1]$: it reflects controlled covariation, rather than a probability or a variance share. Monthly aggregation, conditioning and horizon trimming can change that covariation even when the event-level source construction is orthogonal.

## 5 Uncertainty in the adjustment gap

The finite-sample identity does not imply that its mixture weight should be treated as fixed for every inferential target. In the population projection, $w_{c,h}$ is a moment ratio of the rate and information residuals after projection on $Z$. Its estimated value varies with the regressor sample. Holding the supplied factor and selected rotation fixed is a different decision from holding realised regressor paths, or their mixture weight, fixed.

Under regular joint asymptotic linearity, the gap's influence satisfies

$$\psi_{\delta,t} = w_c\psi_{\theta,t} + \theta\psi_{w,t}, \tag{11}$$

with horizon subscripts suppressed. Thus its variance includes component-response variation, weight variation and their covariance. Appendix A.2 derives each term, including the effect of residualising on estimated controls. Using only $|\widehat{w}_c|\,\mathrm{SE}(\widehat{\theta})$ omits the latter two terms under alternatives; it does not establish conditional coverage by itself. At $w_c = 0$ but $\theta \neq 0$, weight variation is the leading first-order term. At $w_c = \theta = 0$, the product gradient vanishes and ordinary gap-Wald inference is not generally regular.

I fit the full and reduced regressions with their own residuals, alongside the projection of $c$ on $[r, Z]$ that estimates $w_c$. I retain their joint coefficient contributions and cross-model covariance. The covariance uses Bartlett weights at actual calendar distances, bandwidth

$$L_h = h + \left\lceil 4(n/100)^{2/9} \right\rceil, \tag{12}$$

and one common multiplier $n/(n - k_F)$, where $k_F$ is the full model's column count. At the primary horizon, $L_{12} = 18$. The bandwidth grows with sample size; the correction is a finite-sample convention, rather than a guarantee of unbiased covariance or exact coverage. I use normal critical values and no prewhitening.

The procedure requires regular regressor moments, sufficient higher moments, a suitable weak-dependence central limit theorem and consistent long-run covariance estimation. The periodic calendar controls require the corresponding periodic mean or moment limits to be accommodated. Structural

breaks, arbitrary trends and influential observations are not made harmless by naming HAC. Serially uncorrelated surprises also do not establish serially uncorrelated projection scores with overlapping outcomes. The lag-augmentation result of Montiel Olea and Plagborg-Møller (2021) has state-lag and innovation assumptions that my constructed-shock design does not establish automatically.

For the primary comparison I use a conservative product interval. I form individual 97.5 per cent normal intervals for $w_c$ and $\theta$, with critical value $z_{0.9875}$, and take the smallest and largest of their four endpoint products. The Bonferroni union bound gives asymptotic joint coordinate coverage of at least 95 per cent; mapping that rectangle covers their product. Appendix A.3 proves the coverage statement. It avoids requiring an invertible joint covariance or a regular product gradient, while retaining the need for valid, nondegenerate marginal inference. I do not clip weights to $[0, 1]$. The null $\delta_h = 0$ is the union of $w_{c,h} = 0$ and $\theta_h = 0$. The declared rectangle is deliberately conservative; its non-rejection does not imply agreement among all valid zero-gap tests. Under a maintained nonzero weight, a restricted-null test of $\theta_h = 0$ has a different justification. The preselected product interval remains primary. Section 6.2 reports supplementary procedures introduced after inspection to assess the zero-product geometry.

Individual response intervals and the normal stacked-gap interval remain pointwise; the latter is a regularity-dependent diagnostic. Neither supplies simultaneous coverage over a response path. Ordinary uncertainty conditions on the supplied factor, rotation, measurement vintage and specification. It does not incorporate their construction or identification-set uncertainty.

# 6 Empirical evidence

## 6.1 The primary twelve-month comparison

Table 2 reports the matched estimates. The policy-component contrast is −2.29, with a contextual normal interval $[-4.18, -0.40]$. The underlying sector estimates are −3.47 for durable production and −1.18 for non-durable production. Their covariance is retained in the relative response, rather than judging it from separate sector significance. The information-component contrast is 0.66, with an interval including zero. These are cumulative log-change responses per the declared ten-basis-point factor contribution.

Table 2: Primary twelve-month projections

| Quantity | Estimate | 95% interval | Basis |
|---|---|---|---|
| Durable policy response | −3.47 | $[-5.63, -1.31]$ | Pointwise normal |
| Non-durable policy response | −1.18 | $[-2.14, -0.21]$ | Pointwise normal |
| Relative policy response | −2.29 | $[-4.18, -0.40]$ | Pointwise normal |
| Relative information response | 0.66 | $[-1.07, 2.39]$ | Pointwise normal |
| Recomposed-rate response | −1.03 | $[-2.24, 0.17]$ | Pointwise normal |
| Adjustment gap | 1.26 | $[-0.11, 3.31]$ | Conservative product |

Origins: January 2000–August 2025, $n = 308$. Responses are percentage log-change points per ten-basis-point normalised factor contribution. The full model has 74 columns; Bartlett bandwidth is 18. The gap uses the prespecified product interval. Other intervals are individual normal approximations and do not give simultaneous coverage. The underlying sector coefficients retain their joint covariance.

The undivided-surprise coefficient is −1.03 and its interval includes zero. Its point response is 1.26 percentage log points closer to zero than the policy coefficient, in the same cumulative relative-production units. This is the difference between two projection slopes, rather than two identical policy-rate interventions. The primary conservative bounds $[-0.11, 3.31]$ also include zero. The prespecified conservative procedure therefore does not reject the zero-gap hypothesis. The positive point estimate suggests a smaller relative contraction when the contributions are combined, yet the interval accommodates both a small negative difference and a substantial positive difference. Non-rejection is not evidence that the adjustment has no economically relevant effect.

The decomposition attributes the point gap to an information weight of 0.427 and component-response difference of 2.951. Their product is 1.261. The 97.5 per cent coordinate bounds are $[0.312, 0.542]$ for

the weight and $[-0.198, 6.100]$ for the response difference; the latter permits the negative endpoint of the primary product bounds. This identifies how the coefficient comparison is assembled, rather than measuring the fraction of announcements containing information or identifying a distinct expectations channel. The normal component-difference interval alone cannot establish a nonzero product, and differing separate significance labels do not test the gap.

### 6.2 Supplementary inference on the zero-gap restriction

After observing the primary estimates and preliminary test audit, I recorded a follow-up for all six declared twelve-month cases. I examine an intersection-union point-null test and product bounds from a covariance-aware joint region, using the retained coefficients and covariance without additional outcome regressions. Appendix D gives the proofs and all numerical cases. The primary interval remains unchanged; the follow-up was informed by the preliminary results and is not prospective registration.

The primary intersection-union test gives $p = 0.036$, a nominal 5 per cent rejection of the zero-product restriction. Its joint-ellipse product bounds are $[-0.19, 3.18]$. Every joint-region interval includes zero. The poor-man and pre-pandemic tests do not reject, with $p = 0.174$ and $p = 0.414$. Table D.1 reports all six cases without selection.

The point-null test and the joint region concern the same case-specific population gap, but calibrate uncertainty differently. Their disagreement clarifies the conservatism of the primary procedure rather than establishing an economically important minimum gap. Nominal rejection is supplementary evidence for the selected decomposition; construction sensitivity, earlier-period imprecision and the common marginal-inference assumptions limit a stronger transmission conclusion.

### 6.3 The dynamic comparison

Figure 1 presents horizons zero to twenty-four on 296 common origins, January 2000 to August 2024. The common sample avoids letting changing origin composition alone explain differences across horizons.[10] Its twelve-month point gap is 1.21, with conservative bounds $[-0.20, 3.27]$; it must not replace the primary estimate based on 308 origins.

The policy point estimate is negative and the gap point estimate positive throughout the displayed path. The profiles are irregular, and every shaded interval is pointwise. Selected horizons with apparently stronger separation cannot establish path-wide significance or replace the fixed twelve-month comparison. The plot documents the estimated dynamics and their precision on this sample, rather than identifying an invariant timing mechanism across policy regimes.

### 6.4 Declared sensitivity and influence

Figure 2 reports every declared twelve-month gap comparison; Appendix Table B.2 retains the numerical companion results. Six lags and the alternative poor-man decomposition use the same 308 origins. They give gaps of 1.17 and 1.23, respectively, both with conservative intervals including zero. The poor-man comparison is less precise and replaces both current and lagged components. Six lags instead changes the conditioning set. Matched dates therefore remove sample changes while leaving these distinct design changes. The poor-man construction check is not a full confidence analysis over all admissible rotations.

The pre-pandemic comparison requires every production endpoint to occur by December 2019, yielding 228 origins through December 2018. Its gap is smaller, 0.66, and its information coefficient changes sign to $-0.62$. Its policy contrast remains negative, but the interval $[-4.28, 0.11]$ includes zero. The undivided coefficient's interval instead excludes zero. These separate patterns remain insufficient to establish a difference. Its positive gap despite the negative information contrast illustrates that the response ranking, rather than relative information-induced expansion, drives the point comparison. The period change

[10] At $h = 0$, the outcome covers the whole announcement month, including production before the announcement; it is not an instantaneous event-window response.

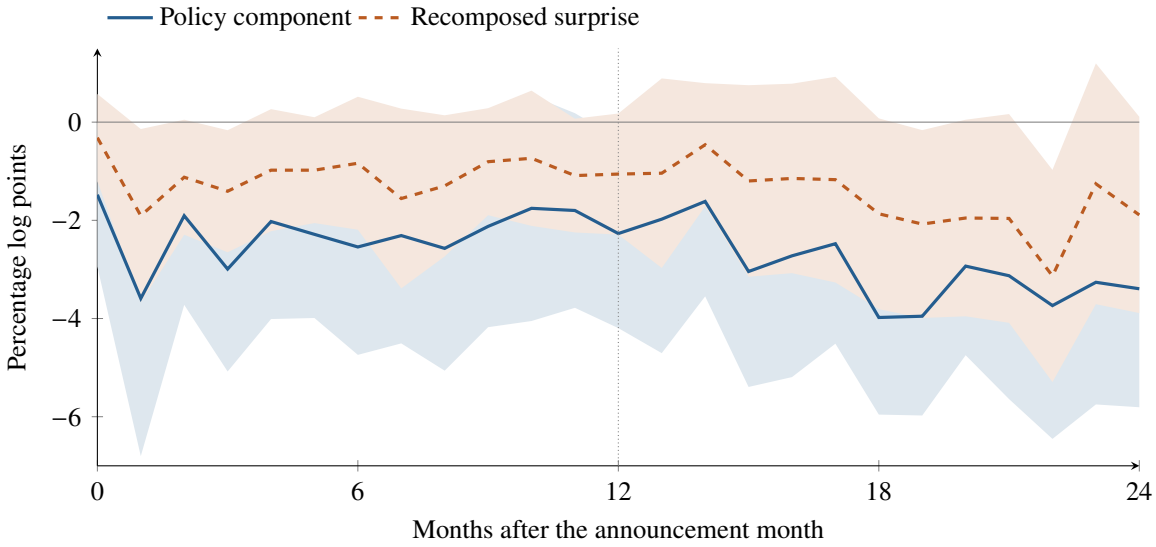



Figure 1: Matched response paths. Coefficients are percentage log-change points per ten-basis-point factor contribution. Shading gives 95 per cent pointwise normal HAC intervals, not simultaneous bands. All horizons use 296 common origins, January 2000–August 2024; the primary twelve-month estimate uses the longer 308-origin sample. Figure 2 shows the directly assessed primary gap; overlap of these individual response bands is not a test of that gap.

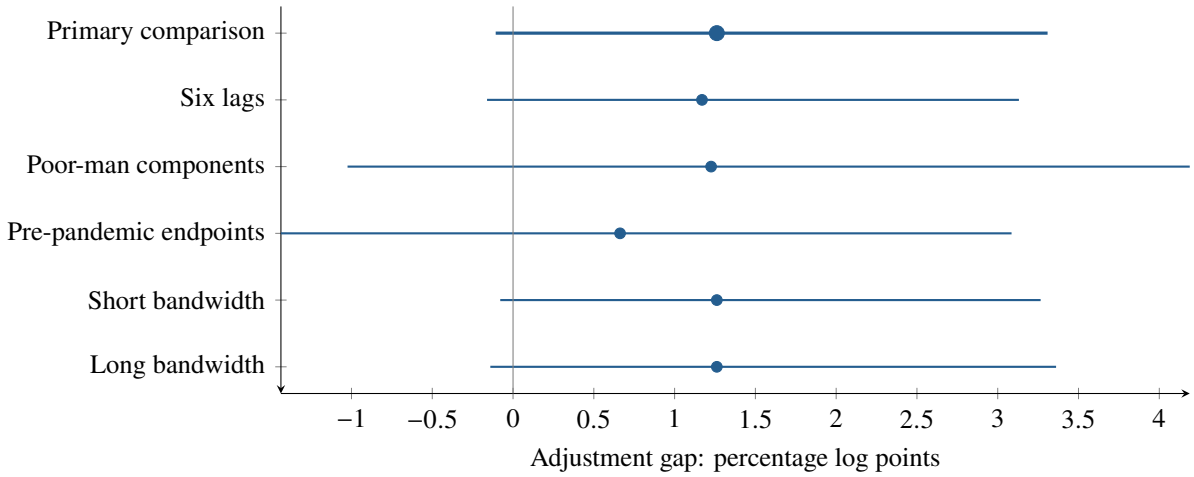



Figure 2: Declared twelve-month adjustment-gap checks. Points show the recomposed-rate coefficient less the policy-component coefficient per ten-basis-point factor contribution. Lines are the conservative 95 per cent product intervals. The larger point and thicker line identify the primary comparison. All samples have 308 origins except the pre-pandemic check, with 228. Every interval includes zero.

also alters policy composition and the historical projection target; I cannot attribute the change solely to pandemic disturbances.

Shorter and longer Bartlett bandwidths, 15 and 24, leave the primary coefficients unchanged. Their conservative gap bounds still include zero. Thus the declared checks retain a positive point gap without resolving its primary imprecision. I preserve this pattern rather than treating the most favourable interval as the result.

Single-origin deletions leave the primary gap positive, but omitting December 2015 changes it by 0.38, approximately thirty per cent of its original value. March, April and May 2020 also appear among the most influential origins. Maximum primary leverage is 0.565. The column-scaled condition number is about 49.1, and the primary control, full and reduced ranks are 72, 74 and 73. These diagnostics show no numerical rank collapse, while highlighting the parameter count and influence burden. With 74 columns in 308 origins and concentrated leverage, nominal coverage remains an assumption-dependent approximation. The common covariance multiplier $308/234$ does not correct leverage-dependent residual shrinkage or establish finite-sample size.[11]

### 6.5 An illustration of omitted variance terms

Multiplying the component-difference standard error by the realised weight gives a standard error of 0.6005 and an illustrative normal interval $[0.08, 2.44]$. Retaining weight variation and cross-model covariance gives a stacked standard error of 0.6749 and a regularity-dependent normal diagnostic interval $[-0.06, 2.58]$. The primary conservative interval remains $[-0.11, 3.31]$.

The three estimated sampling-variance contributions are approximately 0.3606 from component-response uncertainty, 0.0229 from weight variation and 0.0719 from their cross covariance, summing to 0.4554. Direct weight variation contributes about 5.0 per cent and its cross term about 15.8 per cent of the total; together they are the 20.8 per cent omitted by the illustrative arithmetic. Their sign and importance are sample-specific, rather than a universal upward correction. A valid fixed-design analysis requires its own stochastic model. Dropping terms from the marginal random-design variance does not validate such a conditional interval. It cannot substitute for the locked population comparison. This arithmetic illustration applies the established identity to the retained evidence; it neither changes the primary model nor supplies a new theoretical inference result.

## 7 Economic interpretation and limitations

Both consumer-goods groups have negative primary policy-component point responses, with durable production more negative. This is compatible with the conditional user-cost mechanism, but the production indices and nominal announcement units do not identify its household elasticity. The more direct finding concerns response ranking. The information contrast of 0.66 exceeds the policy contrast of $-2.29$; their difference of 2.95, multiplied by the residualised weight 0.427, gives the 1.26 adjustment point. Recomposition therefore weakens the estimated relative contraction without establishing that information news expands either sector. Within the declared 97.5 per cent coordinate bounds, the weight is above zero but the response difference remains unresolved, generating the primary gap interval's inclusion of zero. This locates the imprecision in the component ranking under that conservative construction, while weight variation and cross covariance still enter the population variance.

The difference outcome focuses on heterogeneity but does not automatically remove macroeconomic confounding. An energy-price, foreign-demand or supply disturbance with unequal loadings on the two groups can survive subtraction. Controlling the supplied information component does not demonstrate that the remaining policy component is exogenous. Likewise, an unexpectedly strong systematic response to public news can contaminate the policy-labelled contribution while retaining conventional rate–equity

[11] A single-origin deletion removes one projection row. It does not remove that announcement from every subsequent control history or remove its outcome months from other overlapping rows. I retain it as an influence diagnostic without automatic exclusions.

signs (Bauer and Swanson, 2023a). The maintained announcement interpretation is therefore economically substantive, beyond the numerical sign restrictions and regression identities.

The production allocation also constrains what the estimate can explain. A relative movement in these indices is not a household-expenditure elasticity, a motor-vehicle response or a change in the physical share of durable goods. Inventory, export demand, financing and demand postponement are possible mechanisms. The regressions do not separate them. The point estimates should be read in the source's normalised factor units, not translated into the effect of an actual policy-rate step without an additional normalisation and identification argument.

Finally, the finite sample averages over unconventional-policy periods, later tightening and pandemic states. A pooled linear coefficient need not describe an invariant structural response. Revised historical controls are not the real-time information set; the selected source factor and rotation also use their own construction sample. Independent numerical agreement validates implementation, not these economic assumptions. The combination of imprecise gaps, influential origins and unpropagated identification uncertainty limits a stronger causal claim.

# 8 Conclusion

I find a more negative durable-minus-non-durable production point response when the current policy contribution is separated from the recomposed ECB announcement surprise. The positive twelve-month adjustment point reflects the ranking of the component responses and their controlled announcement weight. Its primary conservative interval and every declared sensitivity interval include zero, so the declared procedure does not resolve the population adjustment. The pre-pandemic comparison also leaves the companion policy contrast imprecise. The supplementary intersection-union test gives nominal evidence against a zero gap for the selected construction, but the joint-region product bounds include zero and the alternative construction and earlier-period tests do not reject.

Economically, combining announcement contributions can weaken an estimated durability contrast without requiring a positive relative information response. The stock-flow benchmark illustrates user-cost and demand mechanisms, but relative prices, inventories, exports and financing prevent reading the production estimates as an identified household mechanism. The measured responses also depend on the mapping from supplied contributions to latent innovations. The study therefore supports a bounded conclusion: announcement measurement changes these point projections, while the declared uncertainty and maintained causal assumptions limit a stronger transmission claim.

# A Projection identities and uncertainty

The following derivations apply established partial-regression and moment-equation algebra to my matched comparison. Ding (2021, Sections 2–3 and Appendix A) provides the relevant published partial-regression and covariance results. I hold the supplied shock construction fixed while allowing sampling variation in the regressors and their projection weights.

## A.1 The exact sample comparison

Fix a horizon and suppress its subscript. Let $Y, m, c \in \mathbb{R}^n$ refer to identical observations, with $r = m + c$ in common units. The $n \times q$ control matrix $Z$ includes an intercept. I require $\operatorname{rank}(Z) = q$, $\operatorname{rank}[Z, m, c] = q + 2$ and $n > q + 2$. Define

$$M_Z = I - Z(Z^\top Z)^{-1} Z^\top, \qquad y = M_Z Y,$$
$$a = M_Z m, \qquad b = M_Z c, \qquad s = a + b. \tag{A.1}$$

Thus $M_Z^\top = M_Z$, $M_Z^2 = M_Z$ and $M_Z Z = 0$. The residualised matrix $A = [a, b]$ has rank two, so $A^\top A$ is positive definite and $s^\top s > 0$.

The identity stated in Proposition 1 is derived as follows.

*Proof of Proposition 1.* For fixed $\beta = (\beta_m, \beta_c)^\top$, minimising the full criterion over its nuisance coefficient gives

$$\widehat{\eta}(\beta) = (Z^\top Z)^{-1} Z^\top (Y - m\beta_m - c\beta_c), \tag{A.2}$$

$$\min_{\eta} \|Y - m\beta_m - c\beta_c - Z\eta\|^2 = \|y - A\beta\|^2. \tag{A.3}$$

Consequently $\widehat{\beta} = (A^\top A)^{-1} A^\top y$. The residual $\widehat{e} = y - A\widehat{\beta}$ satisfies $a^\top \widehat{e} = b^\top \widehat{e} = 0$. Applying the same minimisation to the reduced regression gives

$$\widehat{\gamma} = \frac{s^\top y}{s^\top s} = \frac{s^\top a}{s^\top s}\widehat{\beta}_m + \frac{s^\top b}{s^\top s}\widehat{\beta}_c, \qquad \frac{s^\top a + s^\top b}{s^\top s} = 1. \tag{A.4}$$

Subtracting $\widehat{\beta}_m$ proves (10). □

This is an identity for every $Y$, without probabilistic assumptions. It does not survive different rows, controls, scaling or a different definition of $r$. Expanding its weight shows why it need not be a probability or variance share:

$$\widehat{w}_c = \frac{b^\top b + a^\top b}{a^\top a + b^\top b + 2a^\top b}. \tag{A.5}$$

Negative weights or weights above one are possible. If $\widehat{w}_c = 0$, the sample gap is zero for every outcome although $\widehat{\theta}$ need not vanish. Event-level orthogonality also does not imply controlled monthly orthogonality: for an aggregation matrix $H$, the relevant inner product is $m_e^\top H^\top M_Z H c_e$, rather than $m_e^\top c_e$.

Since $m = r - c$, the full regression can instead be written $Y = \beta_m r + \theta c + Z\eta + e$. Its first two coefficients are the nonsingular transformation $T\beta$, where $T = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$. Fitted values and residuals are unchanged, and a consistently computed covariance transforms as $\widehat{V}_{\beta_m,\theta} = T\widehat{V}_{\beta_m,\beta_c}T^\top$.

## A.2 Population weights and their influence

Let the corresponding population projections exist with nonsingular moment matrices. Expectations below use the stable population moments, or their periodically structured limits, required by the design. A tilde denotes population linear residualisation on $Z$. Write $\widetilde{r} = r - Z^\top \pi_r$ and $\widetilde{c} = c - Z^\top \pi_c$, where $\mathbb{E}[Z\widetilde{r}] = \mathbb{E}[Z\widetilde{c}] = 0$. Put

$$B = \mathbb{E}[\widetilde{r}^2] > 0, \qquad C = \mathbb{E}[\widetilde{r}\widetilde{c}], \qquad w_c = C/B, \qquad \theta = \beta_c - \beta_m. \tag{A.6}$$

For the full residual $e$, the projection normal equations give $\mathbb{E}[Ze] = \mathbb{E}[me] = \mathbb{E}[ce] = 0$. Therefore $\gamma = B^{-1}\mathbb{E}[\widetilde{r}\widetilde{Y}] = \beta_m + w_c\theta$. The reduced residual consequently differs from the full residual:

$$\widetilde{Y} = \beta_m \widetilde{r} + \theta\widetilde{c} + e, \qquad v = e + \theta(\widetilde{c} - w_c\widetilde{r}). \tag{A.7}$$

The nuisance projections do not add first-order terms to $C$ or $B$, because

$$\partial_{\pi_r} C = -\mathbb{E}[Z\widetilde{c}] = 0, \qquad \partial_{\pi_c} C = -\mathbb{E}[Z\widetilde{r}] = 0, \qquad \partial_{\pi_r} B = -2\mathbb{E}[Z\widetilde{r}] = 0. \tag{A.8}$$

Thus, under regular asymptotic linearity, their moment influences are $\widetilde{r}_t\widetilde{c}_t - C$ and $\widetilde{r}_t^2 - B$. Differentiating their ratio gives

$$\psi_{w,t} = \frac{(\widetilde{r}_t\widetilde{c}_t - C) - w_c(\widetilde{r}_t^2 - B)}{B} = \frac{\widetilde{r}_t(\widetilde{c}_t - w_c\widetilde{r}_t)}{B}. \tag{A.9}$$

For $\xi_t = (\widetilde{m}_t, \widetilde{c}_t)^\top$ and $Q = \mathbb{E}[\xi_t\xi_t^\top]$, the full coefficient influence is $\psi_{\beta,t} = Q^{-1}\xi_t e_t$, while $\psi_{\gamma,t} = \widetilde{r}_t v_t/B$. Define $\psi_{\theta,t} = (-1, 1)\psi_{\beta,t}$. The row identity $(1 - w_c, w_c) = \mathbb{E}[\widetilde{r}\xi^\top]/B$ implies

$$\frac{\widetilde{r}_t e_t}{B} = (1 - w_c, w_c)\psi_{\beta,t} = \psi_{\beta_m,t} + w_c\psi_{\theta,t}, \tag{A.10}$$

$$\psi_{\delta,t} = \psi_{\gamma,t} - \psi_{\beta_m,t} = w_c\psi_{\theta,t} + \theta\psi_{w,t}. \tag{A.11}$$

Hence the long-run variance contains

$$\mathcal{V}_\delta = w_c^2 \mathcal{V}_\theta + \theta^2 \mathcal{V}_w + 2 w_c \theta C_{\theta,w}, \tag{A.12}$$

where $C_{\theta,w}$ is the long-run covariance. A fixed-weight propagation retains only the first term. Under a genuinely fixed design $\mathcal{X}$, the sample identity implies $\mathrm{Var}(\widehat{\delta} \mid \mathcal{X}) = \widehat{w}_c^2 \,\mathrm{Var}(\widehat{\theta} \mid \mathcal{X})$. A marginal random-design HAC variance need not estimate that conditional variance. Conditioning only on an estimated weight is also different from fixing the complete design. If $w_c = 0$ and $\theta \neq 0$, weight variation is instead the leading term. If $\theta = 0$ and $w_c \neq 0$, the weight term vanishes under the null; this does not invalidate a response-difference test under a maintained nonzero weight. If both vanish, the first-order product gradient is zero and ordinary gap-Wald inference is not generally regular.

## A.3 Joint covariance and conservative product bounds

I fit the full, reduced and weight regressions, respectively, with $X_F = [m, c, Z]$, $X_R = [r, Z]$ and $X_W = X_R$, using responses $Y, Y, c$ and each regression's own residual $\widehat{u}_{j,t}$. Let $\ell_{j,t}$ be column $t$ of $L_j = (X_j^\top X_j)^{-1} X_j^\top$. The distinction between the population-scale influence and the numerical coefficient contribution is

$$\widehat{\psi}_{j,t} = \widehat{Q}_j^{-1} x_{j,t} \widehat{u}_{j,t} = n \ell_{j,t} \widehat{u}_{j,t}, \qquad \widehat{Q}_j = X_j^\top X_j / n, \qquad a_{j,t} = \widehat{\psi}_{j,t} / n. \tag{A.13}$$

I compute $L_j$ through the full-rank QR factorisation. Let $a_t$ collect the contributions for $\beta_m, \beta_c, \gamma$ and $w_c$. It satisfies $a_{\delta,t} = a_{\gamma,t} - a_{\beta_m,t} = \widehat{w}_c a_{\theta,t} + \widehat{\theta} a_{w,t}$. Its four-dimensional first-order covariance is therefore generally singular; no rank-four inversion is required.

For origin-month ordinals $\tau_i$, the calendar Bartlett covariance is

$$\widehat{V} = \kappa \sum_{i=1}^{n} \sum_{j=1}^{n} K_{L_h}(|\tau_i - \tau_j|) a_i a_j^\top, \qquad \kappa = \frac{n}{n - k_F}, \qquad k_F = q + 2, \tag{A.14}$$

$$K_L(d) = \max\{1 - d/(L+1), 0\}, \qquad L_h = h + \left\lceil 4(n/100)^{2/9} \right\rceil. \tag{A.15}$$

Equivalently, (A.14) uses $\widehat{\psi}_i \widehat{\psi}_j^\top / n^2$. Calendar gaps are retained, not compressed. I apply the same correction to every joint component, without prewhitening. This finite-sample convention does not establish exact coverage. The inference requires a joint score central limit theorem, adequate moments and consistent long-run covariance estimation; lagged growth controls alone do not supply these conditions.

A fixed linear transformation of $\widehat{V}$ yields the covariance of $(\widehat{\beta}_m, \widehat{\delta}, \widehat{w}_c, \widehat{\theta})$, retaining its cross-model blocks. Likewise, shared sector regressions obey $\widehat{\beta}_m = \widehat{\beta}_{D,m} - \widehat{\beta}_{N,m}$ and

$$\widehat{\mathrm{Var}}(\widehat{\beta}_m) = \widehat{V}_{D,m} + \widehat{V}_{N,m} - 2\widehat{\mathrm{Cov}}(\widehat{\beta}_{D,m}, \widehat{\beta}_{N,m}). \tag{A.16}$$

This covariance cannot be inferred from separate significance labels.

**Proposition 2** (Conservative product interval)**.** *Suppose $I_w = [w_-, w_+]$ and $I_\theta = [\theta_-, \theta_+]$ each cover their population coordinate with asymptotic probability at least $1 - \alpha/2$. Then*

$$I_\delta = \left[ \min_{s,t \in \{-,+\}} w_s \theta_t, \max_{s,t \in \{-,+\}} w_s \theta_t \right] \tag{A.17}$$

*has asymptotic coverage at least $1 - \alpha$ for $\delta = w_c \theta$.*

*Proof.* The product's extrema over the coordinate rectangle occur at its corners, since it is linear in either coordinate when the other is fixed. Therefore

$$\begin{aligned} \Pr\{\delta \in I_\delta\} &\geq \Pr\{w_c \in I_w, \theta \in I_\theta\} \\ &\geq 1 - \Pr\{w_c \notin I_w\} - \Pr\{\theta \notin I_\theta\}. \end{aligned} \tag{A.18}$$

Taking the lower limit gives the claimed coverage without independence or an invertible joint covariance. □

I use normal marginal intervals with critical value $z_{1-\alpha/4}$, so at $\alpha = 0.05$ each coordinate interval is 97.5 per cent. Weights are not restricted to $[0, 1]$. The argument applies even at $w_c = \theta = 0$ provided the coordinates themselves admit valid marginal inference with nondegenerate scalar long-run variances. It does not cure marginal degeneracy. The bounds remain asymptotic and pointwise in the horizon. Neither they nor numerical replication establish announcement exogeneity or incorporate uncertainty in the supplied factor loadings, rotation or vintage.

# B Source reconciliation

Table B.1 distinguishes source coverage from the estimation samples. I retain the 15 September snapshot unchanged alongside the active 10 October snapshot. The production update adds August 2026 and revises July non-durable production from 92.3 to 91.4; other overlapping production observations and the historical HICP request are unchanged. The author-supplied shock CSVs are unchanged between these retrievals.

Table B.1: Coverage of retained source observations

| Source | First | Last | Observations |
|---|---|---|---|
| Production, each grouping | 1991-01 | 2026-08 | 428 |
| Historical all-items HICP | 1996-01 | 2025-12 | 360 |
| Monthly contributions | 1999-01 | 2025-10 | 322 |
| Event contributions | 1999-01-07 | 2025-10-30 | 312 |

The EA-MPD workbook contains 315 dated rows. The update explicitly excludes 13 September 2001, 17 September 2001 and 8 October 2008, leaving exactly 312 retained events. Its code records a joint Fed–ECB US-dollar swap announcement, a joint cut at 17:30 and a joint cut, respectively. There are 277 months with retained events and 45 without. Six event factors are observed zeros with nonzero equity surprises and offsetting nonzero median contributions; they differ from empty-event months. No retained four-maturity rate or equity input is missing after these exclusions. All six monthly fields equal event sums to retained numerical precision, and the uncentred component and median rotation reproduce the supplied eight-decimal values. Exact source URLs, the pinned revision and checksums are retained with the data manifest.

## B.1 Specification record and numerical replay

I recorded the primary specification at 21:30 UTC on 10 October 2026 and committed revision `ad01d98` before the first response execution at 23:07 UTC. The outcome, horizon, samples, components, controls and interval convention remained unchanged. The omitted-term calculation, household benchmark, measurement bridge and descriptive table were added after estimation. The subsequent assessment in Section 6.2 was recorded after primary inspection and preliminary diagnostic arithmetic. These additions supplement the primary procedure without an additional fitted outcome model or primary specification departure.

The six-lag and poor-man comparisons retain the primary origins. The earlier-period check requires $t+h \leq$ December 2019, so its twelve-month origins end in December 2018. The short and long bandwidths are $h + \lceil B/2 \rceil$ and $h + 2B$, respectively, for $B = \lceil 4(n/100)^{2/9} \rceil$. These were declared sensitivities, rather than departures selected from the results. Table B.2 records the numerical comparisons.

An independent base-R reconstruction checks transformations, controls, samples, QR coefficients, joint calendar covariance and intervals for nine cases. Its 351 comparisons agree within approximately $1.04 \times 10^{-13}$. This verifies calculation on the same inputs, not independent empirical replication. The private research repository retains data, code, results and the consulted-source register; copyrighted article texts and the ECB workbook remain separately stored.

Table B.2: Numerical sensitivity comparisons at twelve months

| Specification | $n$ | $\widehat{\beta}_m$ | Normal 95% interval | $\widehat{\delta}$ | Product 95% bounds |
|---|---|---|---|---|---|
| Primary | 308 | −2.29 | [−4.18, −0.40] | 1.26 | [−0.11, 3.31] |
| Six lags, matched | 308 | −2.52 | [−4.62, −0.42] | 1.17 | [−0.16, 3.13] |
| Poor-man components | 308 | −2.27 | [−4.06, −0.47] | 1.23 | [−1.02, 4.19] |
| Pre-pandemic endpoints | 228 | −2.08 | [−4.28, 0.11] | 0.66 | [−1.44, 3.09] |
| Short bandwidth | 308 | −2.29 | [−4.16, −0.43] | 1.26 | [−0.08, 3.27] |
| Long bandwidth | 308 | −2.29 | [−4.25, −0.34] | 1.26 | [−0.14, 3.36] |

All responses use the same ten-basis-point factor units. Six-lag and poor-man comparisons retain the primary origins. Pre-pandemic endpoints end by December 2019. Short and long bandwidths are 15 and 24, versus 18 in the primary regression, leaving its coefficients unchanged. Intervals are pointwise and retain the declared primary uncertainty target.

# C Economic mechanism and causal measurement

## C.1 Durable services, user cost and purchase adjustment

I derive a partial household benchmark using the established stock-service distinction of Mankiw (1982, pp. 418–420) and Barsky et al. (2007, pp. 985–986, 990–992); it is not a new durability theorem. Let $0 < \alpha < 1$, $0 < \beta_H < 1$, $0 < d \leq 1$, $p_t > 0$ and $\rho_t > -1$. Consumption and stock are positive; purchases and bond holdings are interior, resale is allowed, and feasibility and transversality hold. There are no adjustment costs. With the stock law (1), the household chooses consumption, stock and a bond with next-period face-value payoff $B_t$ to maximise

$$\max_{\{C_t, S_t, B_t\}} \quad \mathbb{E}_0 \sum_{t \geq 0} \beta_H^t \{\alpha \log C_t + (1-\alpha) \log S_t\},$$
$$\text{subject to} \quad C_t + p_t X_t + \frac{B_t}{1+\rho_t} = W_t + B_{t-1}, \quad t \geq 0. \tag{C.1}$$

Resources $W_t$ are in non-durable units. The real risk-free return $\rho_t$, depreciation and discount factor refer to the same period. This return differs from the nominal recomposed announcement factor $r_t$.

Let $\lambda_t = \alpha / C_t$. Varying the bond and end-of-period stock gives

$$\frac{\lambda_t}{1+\rho_t} = \beta_H \mathbb{E}_t \lambda_{t+1}, \tag{C.2}$$
$$\lambda_t p_t = \frac{1-\alpha}{S_t} + \beta_H (1-d) \mathbb{E}_t [\lambda_{t+1} p_{t+1}]. \tag{C.3}$$

The stock condition values current services and the retained stock's next-period resale value. Set $M_{t+1} = \beta_H \lambda_{t+1} / \lambda_t$. Dividing (C.3) by $\lambda_t$ proves (2), including positive $q_t$. The bond condition and covariance decomposition then imply

$$q_t = p_t - \frac{1-d}{1+\rho_t} \mathbb{E}_t p_{t+1} - (1-d) \operatorname{Cov}_t (M_{t+1}, p_{t+1}). \tag{C.4}$$

A negative covariance raises user cost relative to the expectation-only expression: resale is less valuable in high-marginal-utility states. Both this risk term and expected prices can change with announcement news.[12]

Under certainty and constant relative prices, $q_t = p(\rho_t + d)/(1+\rho_t)$, requiring $\rho_t + d > 0$. This simplification does not require stationary consumption. Stationary consumption at a benchmark would additionally require $\beta_H(1+\bar{\rho}) = 1$.

[12] This household valuation covariance differs from the observed announcement-time rate–equity covariance; it is not estimated by the production regressions.

At a stationary-stock base, $\bar{X} = d\bar{S} > 0$. Holding inherited stock fixed gives $\mathrm{d}X_t = \mathrm{d}S_t$ and hence

$$\mathrm{d}\log X_t = \frac{1}{d}\mathrm{d}\log S_t = \frac{1}{d}(\mathrm{d}\log C_t - \mathrm{d}\log q_t), \tag{C.5}$$

$$\left.\frac{\partial \log X_t}{\partial \rho_t}\right|_{C_t, S_{t-1}, p} = -\frac{1-d}{d(1+\rho_t)(\rho_t + d)}. \tag{C.6}$$

Subtracting $\mathrm{d}\log C_t$ proves (3). The rate derivative is negative for $d < 1$ and zero at $d = 1$. It is a local impact comparison with positive purchases, not a comparison of steady states holding consumption fixed as rates vary. Zero depreciation, a binding no-resale constraint, a finite movement outside the positive-purchase neighbourhood or adjustment costs can invalidate this derivative. Endogenous consumption and prices instead alter the total equilibrium response. The positive-purchase restriction is material: with current $C_t$ fixed,

$$X_t(q_t) = \bar{S}\left\{\frac{\bar{q}}{q_t} - (1-d)\right\} > 0 \quad \Longleftrightarrow \quad q_t < \frac{\bar{q}}{1-d}, \qquad 0 < d < 1. \tag{C.7}$$

As $d$ approaches zero, the amplification grows but this admissible contraction neighbourhood shrinks; a large finite elasticity does not follow from the local derivative. Exports, inventories, firm financing and index aggregation also separate household purchases from measured production.

### C.2 A conditional bridge from components to causal responses

For a fixed horizon, suppose a stable linear structural relation and a noiseless measurement equation hold:

$$Y_{t,h} = b_h^\top \varepsilon_t + Z_t^\top \eta_h + \nu_{t,h}, \qquad s_t = A\varepsilon_t. \tag{C.8}$$

The constant $A$ is invertible and $Q_s = \mathbb{E}[\widetilde{s}_t \widetilde{s}_t^\top]$ is positive definite. Tildes denote population linear residualisation on the same $Z_t$. Maintain $\mathbb{E}[\widetilde{s}_t \nu_{t,h}] = 0$, which implies orthogonality to the residualised disturbance. Since $\widetilde{\varepsilon}_t = A^{-1}\widetilde{s}_t$,

$$\beta_h = Q_s^{-1}\mathbb{E}[\widetilde{s}_t \widetilde{Y}_{t,h}] = Q_s^{-1} Q_s A^{-\top} b_h = A^{-\top} b_h. \tag{C.9}$$

If $A = \mathrm{diag}(a_m, a_c)$ is known with ordered nonzero scales, then $b_{m,h} = a_m \beta_{m,h}$ and $b_{c,h} = a_c \beta_{c,h}$. A known dense mapping also permits joint recovery $b_h = A^\top \beta_h$; a diagonal mapping is needed to read each supplied coefficient as one separately scaled latent response. An unidentified dense $A$ mixes responses. Orthogonality of observed columns constrains their second moments, not the mapping's diagonality or causal exogeneity. Noise in $s_t$ generally breaks (C.9). A causal interpretation requires the structural relation, invariance and orthogonality in addition to matched regression rows and financial sign restrictions. The supplied point rotation and HAC intervals do not establish these assumptions or cover its entire identification set. Here $b_h$ is a difference of latent sector responses, not an intervention on a physical production ratio.

## D Supplementary zero-product inference

The null is $H_0 = H_w \cup H_\theta$, where the coordinate nulls are $w = 0$ and $\theta = 0$. Let $p_w, p_\theta$ be the existing two-sided normal marginal $p$-values, asymptotically valid under their respective nulls. The established intersection-union principle (Berger and Hsu, 1996, §3, Theorem 1) gives

$$p_{\mathrm{IUT}} = \max\{p_w, p_\theta\}. \tag{D.1}$$

At any parameter point $a \in H_0$, at least one coordinate null $H_j$ is true, so

$$\limsup_n \Pr_a(p_{\mathrm{IUT}} \le \alpha) \le \limsup_n \Pr_a(p_j \le \alpha) \le \alpha. \tag{D.2}$$

No independence is required. This establishes pointwise asymptotic level under valid coordinate inference, not finite-sample or uniform validity. With the declared normal marginals, primary rectangle exclusion of zero requires both $p$-values below 0.025. Its rejection is therefore nested within the supplementary 5 per cent test. Selection between those nested gates alone does not increase rejection beyond the latter gate; broader post-estimation selection is not justified by this argument.

For $a = (w, \theta)^\top$, assume a joint Gaussian limit and positive-definite estimated covariance $\widehat{V}_a$. The first-order 95 per cent joint region and its product image are

$$\mathcal{E}_{.95} = \{a : (a - \widehat{a})^\top \widehat{V}_a^{-1} (a - \widehat{a}) \leq \chi^2_{2,.95}\}, \tag{D.3}$$

$$I_{\delta,\mathcal{E}} = \left[ \min_{a \in \mathcal{E}_{.95}} w\theta, \max_{a \in \mathcal{E}_{.95}} w\theta \right]. \tag{D.4}$$

Joint coordinate coverage implies product coverage. Writing $\widehat{V}_a = LL^\top$ parameterises its boundary by $\widehat{a} + \sqrt{\chi^2_{2,.95}} L(\cos\phi, \sin\phi)^\top$. The product's stationary condition becomes a quartic in $\tan(\phi/2)$. I evaluate all real roots and the omitted point $\phi = \pi$; the only interior stationary point is the origin, a saddle. Independent optimisation checks the bounds.

The zero-product set is the union of the coordinate axes. Profiling the quadratic distance over each axis gives

$$0 \in I_{\delta,\mathcal{E}} \quad \Longleftrightarrow \quad \min \left\{ \frac{\widehat{w}^2}{\widehat{V}_{ww}}, \frac{\widehat{\theta}^2}{\widehat{V}_{\theta\theta}} \right\} \leq \chi^2_{2,.95}. \tag{D.5}$$

The primary response-difference statistic squared is approximately 4.41, below the ellipse threshold 5.99, explaining its inclusion of zero. These procedures summarise the same gap without repairing inaccurate score inference, construction uncertainty or announcement endogeneity.

Table D.1: Post-estimation twelve-month zero-gap assessment

| Specification | Intersection-union $p$ | Joint-ellipse product 95% bounds |
|---|---|---|
| Primary | 0.036 | $[-0.19, 3.18]$ |
| Six lags | 0.043 | $[-0.24, 2.82]$ |
| Poor-man components | 0.174 | $[-1.17, 3.03]$ |
| Pre-pandemic endpoints | 0.414 | $[-1.25, 2.94]$ |
| Short bandwidth | 0.033 | $[-0.17, 3.12]$ |
| Long bandwidth | 0.039 | $[-0.21, 3.26]$ |

The same six designs and ten-basis-point factor units are retained. Nominal levels require valid marginal or joint asymptotic inference. The procedures were recorded after primary inspection and preliminary diagnostic arithmetic; they do not replace the primary product bounds. No confidence claim across all six designs is made.

The retained production endpoint, August 2026, limits the twelve-month origins to August 2025. The fixed shocks extend through October 2025, but the next two complete origins require subsequent production endpoints. Updating announcements or changing countries and production categories cannot supply an independent replication of this defined German contrast. The follow-up therefore uses existing evidence; it makes no significance-driven sample extension or stopping decision.

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